

Mehrdad Alemzadeh

Email: Zade.Mehrdad@gmail.com

Location: Toronto, Ontario | Canadian Citizen

Links: [LinkedIn](#) | [GitHub](#) | [Website](#)

Summary

Senior solution architect and high-performing full stack software engineer with a proven record of leading and delivering impactful solutions for mission-critical initiatives across multiple teams and departments. Skilled in cloud services and end-to-end SDLC ownership, with expertise in leveraging AI services to add meaningful values. Experienced in setting technical direction, mentoring engineers, and driving cross-functional collaboration at scale. Expert problem-solving skills and consistent ability to deliver cost-effective, high-quality outcomes.

Currently serving as a multi-faceted lead at Aletha Corp., navigating complex client engagements as a solution architect and tech lead. I have been involved in departments such as Wealth, Mortgage, Payments, Commercial banking, Fraud, HR, Cloud, DevOps and AI-Innovations. Complementing corporate expertise with an entrepreneurial background as a founder of multiple live software solutions, demonstrating a unique blend of technical rigor and product vision. I have over a decade of experience with AI from building, training and testing models to delivering dozens of production grade solutions.

Honors

- Among Top 3 Enterprise AI Solutions — CIBC's Dragon's Den Innovation Challenge (*December 2025*)
 - Among Top 20 Developers at BNS (with only 6 month worth of contributions) — BlueOptima (*June 2023*)
 - Ranked #1 in T-Factor at TCS for 3 years (*2018–2021*)
 - 4-Year Academic Scholarship — McMaster University
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Education

- Doctor of Philosophy (PhD), Computer Science
 - McMaster University (*Completed Thesis 2016*)
 - Machine Learning Models for Medical Image Analysis
- Master of Science (MSc), Information Systems
 - Statistical Feature Extraction and Neural Network Training
- Bachelor of Science (BSc), Software Engineering
 - Image Processing

Skills

Category	Details
Strategy & Stakeholders	Executive-level roadmap planning · Multi-million-dollar migration strategy · Cross-functional stakeholder alignment · Cost reduction & optimization · Vendor engagement & architecture governance
Leadership & Delivery	Onsite/offshore team management · Hiring & technical interviewing · Developer mentoring & onboarding · Delivery planning & capacity management · SDLC ownership · ITIL release management
Solution Architecture	Distributed systems design · High availability & disaster recovery · Scalability . Performance . Micro-services & API design · Event-driven architecture · Enterprise integration (REST / SOAP / MQ) · AI Services . Cloud Services . Authentication . Security & Privacy . Modernization . Cost Reduction . Monitoring & log management . Data
AI & Data Engineering	RAG / LLM integration · Prompt engineering · Agentic workflows · ML pipelines · Databricks · Azure OpenAI · Form Recognizer . Token aware solutions . Risk and Security
Languages & Frameworks	Go (Chi), Spring Boot, .NET, FastAPI / Django / Flask (Python), Express.js, Next.js, React, Angular, Swift / SwiftUI / SwiftData, AVFoundation, iOS, Android, JavaScript, TypeScript, Chrome Extensions, Tailwind CSS, Bootstrap, HTML, CSS
LLMs & Agentic AI	Anthropic Claude, OpenAI ChatGPT, Grok, Meta Llama, DeepSeek, GitHub Copilot, Ollama, Eliza, Azure AI Foundry, AWS Bedrock, Selenium, Playwright
Cloud & Infrastructure	Azure, AWS, GCP · Azure OpenShift · Kafka · Docker · Kubernetes · Azure Bicep · Ansible · Firebase Cloud Functions
Real-Time & Communications	WebRTC · LiveKit · CallKit · PushKit · APNs / VoIP Push · Firebase Cloud Messaging (FCM) · Firestore real-time listeners
DevOps & CI/CD	GitHub Actions, Azure DevOps Pipelines, Jenkins, Makefile
Security	OAuth 2.0, JWT, API Keys, SSL Certificates, Sessions, Cookies, Apigee, API Proxies, RBAC, Azure Entra ID, Sign in with Apple, iOS Keychain
Databases	PostgreSQL, MS SQL / SQL MI, Oracle, MySQL, MongoDB, Firebase / Firestore
Observability	Splunk, Dynatrace, Azure Monitor, Application Insights, Elasticsearch / ELK
Blockchain	Ethereum, Truffle Framework, Solidity (Smart Contracts), Ganache

Work Experience

CIBC — Commercial Banking & Payments | *Solution Architect (via Aletha)*

June 2025 – Present

Project 1 — Safer Payment Canada Migration (IBM to CIBC Azure)

- Drove design, POCs, application launch, and cost reduction strategies for a multi-million-dollar infrastructure migration.
- Planned and executed a migration roadmap targeting millions of dollars in annual cost savings.
- Engineered network configuration across multiple VNets with NSG rules for secure segmentation.
- Configured F5 load balancers for local high availability and global disaster recovery.
- Implemented APIM, RBAC, MSI, and SSL configurations for authentication, authorization, and security.
- Millions of dollars savings for the client with the new infra architecture and support cost

Project 2 — FAE Pinot US Payment Fraud Solution

- Lead architect building infrastructure to support fraud detection across multiple payment touchpoints.
- Architected highly available distributed systems across Azure and hybrid infrastructures using Kafka and Azure OpenShift.
- Embedded observability through Splunk, Dynatrace, and Azure Monitor for proactive anomaly detection.
- Delivered cost-effective, scalable solutions balancing regional failover with optimized service sizing.
- Full tech lead for end-to-end delivery.
- Saved a million dollar with a scalable solution for the client to host several future data sources
- Azure-based solution with internal and external integrations via SOAP and REST APIs, VNet peering, VPN Tunnel, SSL, and SPNs.
- Integrations with SMTP, ServiceNow, CyberArk, Splunk, FeedHub, and Azure Monitor.

Project 3 — CMO Mobile Migration & Modernization

- Lead solution architect migrating a legacy mobile and web application from CGI infrastructure to CIBC Azure, including new features and functionality.
- Supporting 3,000+ users at go-live.

Project 4 — Other involvements

- Due to successful delivery of Safer Payment project I got involved in a second version of this app through an independent infra on Azure as a tech lead and solution architect to further reduce cost on the client by introducing similar infra cost optimizations from the first project
- Similarly, I got engaged in another FAE project for similar purposes
- These projects create over a dozen sub-projects such as SPUS-Pinot, SPC-EPE, SPC-Edge, and etc that I played a major role as a solution architect and hands-on delivery lead

Technologies: Azure OpenShift, Spring Boot, React, APIM, PostgreSQL, Kafka, Oracle, RSA, VPN Tunnel, Akamai, F5, Azure Blob, AKV, HashiCorp Vault, Dynatrace, Splunk, GitHub Actions, CyberArk, Google Firebase (FCM), VMs, SQL MI

CIBC — Enterprise Wires Modernization | *Senior Consultant (via Aletha)*

January 2024 – June 2025

- Owned end-to-end ODS solution architecture, DevOps, design, capacity planning, and delivery.
- Delivered **10x performance improvement** on the AMH parser application through hands-on development and architectural redesign.
- Boosted SQL MI database performance on Azure by **25%** by introducing a write-DB to segregate read/write indexes.
- Achieved **8x throughput improvement** on Swift MT-to-MX completeness validation.
- Optimized and generalized ADF pipelines for file transfers across upstream and downstream systems.
- Modernized analytics strategy through Databricks and Generative AI integrations.
- Resolved security deviations and led Tier 1 (high resiliency / high availability) migration planning and design roadmap.

Technologies: Azure, Spring Boot, Azure Function Apps, Azure Data Factory, SQL MI, Application Insights, Azure Monitor

ONEX | *Tech Lead & Architect (via Aletha)*

July 2023 – December 2024

- Architected and delivered HRIS go-live; integrated Hibob with 10+ services including Shareworks, Carrot, Optum, iA, OCA, ActivPayroll, Active Directory, and Business Central.
- Extended an open-source Microsoft project into a production multi-index RAG solution — integrating Azure OpenAI, Microsoft Search, and Form Recognizer — **saving ~\$300K/year** in legal document review costs.
- Designed secure RBAC and Azure Entra ID access models for the AI-powered RAG/chatbot system, ensuring authorized-only data access.
- Engineered an invoice/expense integration service between Coupa and Business Central using Azure Function Apps.

Technologies: Azure, Azure Function Apps, .NET, Spring Boot, Flask, Azure OpenAI, Microsoft Search, Form Recognizer, React

AutoSavvy | *Application Architect (via Aletha)*

October 2023 – 2024

- Optimized workflows and designed scalability strategies to improve platform throughput.
- Designed and normalized database schemas with rigorous validation procedures.

Technologies: SQL SaaS, Azure Data Factory

Scotiabank — Wealth Management | *Tech Lead (via Aletha)*

March 2022 – July 2023

- Led as principal architect and technical lead, owning project delivery and aligning solutions with business goals through close stakeholder collaboration.
- Designed and delivered five key applications: Fixed Income API, Remediation Application, Real-Time XML Service, ADPR App, and Mediator API.
- Co-authored a 5-year modernization roadmap for the Wealth department, defining phased delivery strategies, platform integrations, and a GCP migration path.

Technologies: Gradle/Spring Boot, Maven/Spring Batch, React, MS SQL SaaS, Oracle, REST & SOAP APIs, JAXB/WSDL/XSD/XML, Azure, WAM OAuth2, Apigee, Splunk, Dynatrace

Aletha — Side Projects | *Sr. Application Architect*

February 2022 – Present

- AI-based project effort estimator with chatbot generating high-level architecture, deliverables, and PERT-based effort breakdowns.
 - Interview agent full-stack solution for first-round candidate screening.
 - Custom full-stack Azure cost dashboard by resource group and service breakdown.
 - SuiteCRM deployment to Azure.
 - Timesheet app integrated with Azure Entra ID (Flask, React, MSSQL) — *Jan 2022*
 - Chrome extension integrated with OpenAI for page summarization — *Jan 2023*
 - Education app integrated with OpenAI; Dockerized FastAPI and PostgreSQL — *Feb 2024*
 - Conducting interview sessions for hiring and mentoring team on AI trends.
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CIBC — Personal Banking Product & Technology | *Tech Lead (via TCS)*

April 2014 – February 2022

- Directed onsite/offshore teams of 20+ developers, BAs, and QA engineers across programs spanning mortgage securitization, mortgage sale optimization, Azure migration, archive automation, DevOps modernization, log management, ITIL release management, production support, and CMHC/insurer reporting APIs.

Technologies: Java, Spring Boot, Express/Node.js, Python, C#, VB, MS SQL, Git, GitHub, ClearCase, Maven, Jenkins, Ansible, UFT, Elasticsearch, ELK, Splunk, Dynatrace, Jira/Confluence, Docker, Kubernetes, Azure, Power Automate, Tableau, Crystal Reports

St. Joseph's Hospital | *Software Engineer & Researcher*

September 2014 – June 2016

- Partnered with the head of St. Joseph's Imaging Center to automate radiological lung image analysis for disease pattern classification.
 - Built a MATLAB-based classifier achieving **92%+ accuracy** across 10 irregular radiological patterns using neural networks and machine learning.
 - Developed a Python feature extraction tool to identify regions of interest (ROIs), working directly with radiologists to validate clinical relevance.
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Side Projects

i. ****Maester.Work** | Apr 2026 – Present

Designed, built, and deployed a production-grade **AI-powered hiring intelligence platform** as a solo full-stack project — covering product design, architecture, backend, frontend, infrastructure, and billing in a single continuous build.

Platform Overview: A dual-sided SaaS marketplace: hiring managers upload resumes and job descriptions for instant AI-scored fit analysis; job seekers build profiles and let the AI autonomously discover, score, and track matched opportunities across company career pages.

Architecture Highlights

- Stateless Go (Chi) REST API on Cloud Run; zero cold-start overhead at scale-to-zero pricing
- Firebase Auth JWT verification middleware; all Firestore writes restricted to the backend service account — client writes blocked at the security rules layer
- Fully containerized local development stack with Firebase Auth + Firestore emulators via Docker Compose; production mirrors the same topology on GCP
- One-command deployment pipeline: automated GCP provisioning, IAM, Secret Manager, Docker build/push, and Cloud Run deploy via custom shell scripts

Technologies: Go · Chi · Next.js 15 · TypeScript · Tailwind CSS · Anthropic Claude API (Sonnet 4.6, Haiku 4.5) · Firebase Auth · Cloud Firestore · Google Cloud Run · Artifact Registry · Secret Manager · Stripe Checkout · Docker · GCP IAM · Firebase Emulator Suite

ii. **Gist Reader** — iOS App Store (Mar 2026 – Present)

Designed, built, and shipped a production native iOS app as a solo independent project — covering product design, architecture, AI integration, monetization, and App Store submission end-to-end.

What It Does: A book and movie companion app that delivers AI-powered multi-section summaries, real-time multi-source search, text-to-speech playback, and a personal library — built on a polished liquid glass SwiftUI design.

Technologies: Swift · SwiftUI · SwiftData · MVVM · Firebase Auth · Cloud Firestore · Firebase App Check · Google AdMob · OpenAI API · Anthropic Claude API · SSE Streaming · iOS Keychain

iii. **ALO — Real-Time Chat, Voice & Video Calling for iOS** (May 2026 – Present)

Designed, built, and shipped a production-quality native iOS communication app as a solo independent project — covering identity, real-time messaging, WebRTC calling, push infrastructure, and Firebase backend end-to-end.

What It Does: A 1:1 and group messaging app with audio/video calling, real-time text and media exchange, lock-screen call delivery, and a polished liquid-glass SwiftUI interface. Users are identified solely by their Apple ID email — no usernames or passwords.

Technologies: Swift · SwiftUI · MVVM (Observation) · Sign in with Apple · Firebase Auth · Cloud Firestore · Firebase Storage · Firebase Cloud Functions · FCM · LiveKit (WebRTC) · CallKit · PushKit · APNs VoIP · AVFoundation · iOS Keychain

Certificates

Cloud & DevOps: Microsoft Azure AZ-900 · Cloud Architecture · DevOps Foundations · Six Sigma Foundations · Salesforce Integration (SOAP API) · Google Cloud Platform Fundamentals

AI & Machine Learning: Azure Machine Learning for Data Scientists · GitHub Copilot for Project Management · Generative AI for Leaders · Change Management for Generative AI · Neural Networks and Deep Learning ([deeplearning.ai](#)) · Improving Deep Neural Networks ([deeplearning.ai](#)) · Convolutional Neural Networks ([deeplearning.ai](#)) · Sequence Models ([deeplearning.ai](#)) · Machine Learning — Andrew Ng (Stanford) · Machine Learning A–Z: Python & R · Data Science and ML Bootcamp

Programming & Full-Stack: Foundational C# with Microsoft · OOP in Java (University of Helsinki) · Python and Django Full Stack Web Developer · The Web Developer Bootcamp · React — The Complete Guide (Hooks, Router, Redux) · MERN Stack Front to Back · Full Stack: Angular and Java Spring Boot · E-Commerce App with .NET Core and Angular · Blockchain & Cryptocurrency Full Stack

Mobile: iOS 12 & Swift Bootcamp · Android Oreo Developer

Big Data & Analytics: The Ultimate Hands-On Hadoop · Tableau and Crystal Reports (*also referenced in Experience*)

Open Source Contributions

AI-Powered Solutions: x-university (AI auto-generated courses for different age groups) · Agentic-AI (frontend developer integrated with Microsoft Teams) · PortfolioGPT (stock market analysis with LSTM models) · Multi-index ChatGPT solution with privacy management · Book Summarizer Chrome extension · Twitter & Reddit sentiment analysis tools · Azure video-to-text processing pipeline

Full-Stack Development: Interview Agent · Project Effort Estimator · Telegram-based news aggregator · CRM ideathon · Azure custom cost and security dashboard

Enterprise Applications: Flight Reservation System with check-in · Clinical Data Reporting Platform · eCommerce application · Document management system · Location tracking web application

Mobile Applications: Stock price tracker with AI suggestions · Weather forecasting app · Bitcoin price monitor · Bluetooth device finder · Custom Chrome browser with ad-blocker (*Android & iOS*)

Automation & DevOps Tools: macOS app arranger across virtual desktops/monitors/display positions · Windows custom hotkeys to mimic macOS keyboard shortcuts · Automated Yahoo Finance advertisement bot · Video download automation · macOS feature enablement shortcuts · Stock price web scraper · News summarization system

Blockchain Solutions: Full-stack cryptocurrency application · Smart contract development (Solidity) · Blockchain integration with modern web frameworks · Decentralized application (DApp) development · Personal bill/tax tracker

Production Systems: CIBC Commercial Banking modernization · Scotiabank Wealth Management platforms · Enterprise wire processing systems · Multi-million dollar migration projects · High-availability financial applications